ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4	$d = \frac{t_n - t_1}{n-1} = \frac{40}{10}$.
2	**(c)** 9	$a = t_n - (n-1)d = 69 - 60$.
3	**(c)** 59	$5d = 20$ so $d = 4$; then $t_{15} = t_9 + 6d$.
4	**(b)** 182	$2 + 4 + \cdots + 2n = n(n+1)$.
5	**(b)** 196	$1 + 3 + 5 + \cdots + (2n-1) = n^2$.
6	**(b)** 3	$t_5 / t_3 = r^2$.
7	**(a)** 19	These form an AP; count with $n = \frac{l-a}{d} + 1$.
8	**(c)** 70	$t_{20} - t_{10} = 10d$.
9	**(a)** -3	$d = \frac{t_n - t_1}{n-1} = \frac{-24}{8}$.
10	**(d)** 30	$a = t_n - (n-1)d = 16 + 14$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(q-p)d = 30$ (1); $d = 6$ (1).	1+1
12	First 15, last 100; $n = 18$ (1); Sum $= 1035$ (1).	1+1
13	$r^3 = 27$ (1); $r = 3$, $a = 2$ (1).	1+1
14	$(q-p)d = -24$ (1); $d = -3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n = \frac{l-a}{d} + 1 = 25$ (1); $S = \frac{n}{2}(a+l)$ (1); $S = 2000$ (1).	1+1+1
16	$d = 5$ (1); $a = 6$ (1); $t_{20} = 101$ (1).	1+1+1
17	$d = \frac{26-5}{7} = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{20} = 670$ (1).	1+1+1
	OR: $n = \frac{l-a}{d} + 1 = 20$ (1); $S = \frac{n}{2}(a+l)$ (1); $S = 690$ (1).	1+1+1
18	$d = -4$ (1); $a = 30$ (1); $t_{14} = -22$ (1).	1+1+1
	OR: $d = \frac{0-12}{6} = -2$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = -30$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lengths form an AP with $d = -4$ (1); Top rung $= 120 - 10 \times 4 = 80$ cm (1); $S = \frac{n}{2}(a+l)$ (1); $S = \frac{11}{2}(120 + 80)$ (1); Total 1100 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $110/100$ (1); Hence a GP with this common ratio (1); Amount after n years $= P(1 + \frac{r}{100})^n$ (1); $= 10000 \times (1.1)^3$ (1); Amount ₹13310 (1).	1+1+1+1+1
20	Lengths form an AP with $d = -3$ (1); Top rung $= 90 - 14 \times 3 = 48$ cm (1); $S = \frac{n}{2}(a+l)$ (1); $S = \frac{15}{2}(90 + 48)$ (1); Total 1035 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $105/100$ (1); Hence a GP with this common ratio (1); Amount after n years $= P(1 + \frac{r}{100})^n$ (1); $= 20000 \times (1.05)^2$ (1); Amount ₹22050 (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** -8	$5d = -10$ so $d = -2$; then $t_{15} = t_9 + 6d$.
2	**(d)** 306	$2 + 4 + \cdots + 2n = n(n+1)$.
3	**(b)** 361	$1 + 3 + 5 + \cdots + (2n-1) = n^2$.
4	**(b)** 4	$t_5/t_3 = r^2$.
5	**(b)** 19	These form an AP; count with $n = \frac{l-a}{d} + 1$.
6	**(d)** -30	$t_{20} - t_{10} = 10d$.
7	**(c)** 13	$d = \frac{t_n - t_1}{n-1} = \frac{65}{5}$.
8	**(b)** 4	$a = t_n - (n-1)d = 70 - 66$.
9	**(a)** 76	$5d = 25$ so $d = 5$; then $t_{15} = t_9 + 6d$.
10	**(b)** 506	$2 + 4 + \cdots + 2n = n(n+1)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	First 24, last 144; $n = 16$ (1); Sum $= 1344$ (1).	1+1
12	$r^3 = 8$ (1); $r = 2$, $a = 5$ (1).	1+1
13	$(q-p)d = 63$ (1); $d = 7$ (1).	1+1
14	First 36, last 192; $n = 14$ (1); Sum $= 1596$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n = \frac{l-a}{d} + 1 = 31$ (1); $S = \frac{n}{2}(a+l)$ (1); $S = 4588$ (1).	1+1+1
16	$d = 9$ (1); $a = 1$ (1); $t_{25} = 217$ (1).	1+1+1
17	$d = \frac{43-3}{5} = 8$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{12} = 564$ (1).	1+1+1
	OR: $n = \frac{l-a}{d} + 1 = 36$ (1); $S = \frac{n}{2}(a+l)$ (1); $S = 2430$ (1).	1+1+1
18	$d = 8$ (1); $a = -12$ (1); $t_{18} = 124$ (1).	1+1+1
	OR: $d = \frac{27+5}{8} = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{16} = 400$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lengths form an AP with $d = -5$ (1); Top rung $= 100 - 8 \times 5 = 60$ cm (1); $S = \frac{n}{2}(a+l)$ (1); $S = \frac{9}{2}(100 + 60)$ (1); Total 720 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $120/100$ (1); Hence a GP with this common ratio (1); Amount after n years $= P(1 + \frac{r}{100})^n$ (1); $= 50000 \times (1.2)^3$ (1); Amount ₹86400 (1).	1+1+1+1+1
20	Lengths form an AP with $d = -6$ (1); Top rung $= 150 - 12 \times 6 = 78$ cm (1); $S = \frac{n}{2}(a+l)$ (1); $S = \frac{13}{2}(150 + 78)$ (1); Total 1482 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $125/100$ (1); Hence a GP with this common ratio (1); Amount after n years $= P(1 + \frac{r}{100})^n$ (1); $= 8000 \times (1.25)^2$ (1); Amount ₹12500 (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $t_n = a + nd$ instead of $a + (n-1)d$, which makes the first term wrong.
- Forgetting the factor $\frac{n}{2}$ in S_n, or using $n(n+1)$ for the sum of the first n natural numbers (it is $\frac{n(n+1)}{2}$).
- Using $t_n = ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of $S_n - S_{n-1}$.
- Sign errors with a negative common difference or ratio, e.g. $t_5 = 7 + 4 \times (-2) = -1$, not $7 + 4 \times 2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 625	$1+3+5+\cdots+(2n-1)=n^2$.
2	**(b)** 5	$t_5/t_3=r^2$.
3	**(d)** 43	These form an AP; count with $n=\frac{l-a}{d}+1$.
4	**(d)** 50	$t_{20}-t_{10}=10d$.
5	**(b)** 7	$d=\frac{t_n-t_1}{n-1}=\frac{91}{13}$.
6	**(c)** -8	$a=t_n-(n-1)d=76-84$.
7	**(d)** 38	$5d=15$ so $d=3$; then $t_{15}=t_9+6d$.
8	**(b)** 1260	$2+4+\cdots+2n=n(n+1)$.
9	**(c)** 1024	$1+3+5+\cdots+(2n-1)=n^2$.
10	**(d)** 6	$t_5/t_3=r^2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$r^3=64$ (1); $r=4$, $a=3$ (1).	1+1
12	$(q-p)d=28$ (1); $d=4$ (1).	1+1
13	First 42, last 175; $n=20$ (1); Sum $=2170$ (1).	1+1
14	$r^3=125$ (1); $r=5$, $a=1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$n=\frac{l-a}{d}+1=18$ (1); $S=\frac{n}{2}(a+l)$ (1); $S=1755$ (1).	1+1+1
16	$d=3$ (1); $a=22$ (1); $t_{30}=109$ (1).	1+1+1
17	$d=\frac{40-20}{4}=5$ (1); $S_n=\frac{n}{2}[2a+(n-1)d]$ (1); $S_{10}=425$ (1).	1+1+1
	OR: $n=\frac{l-a}{d}+1=27$ (1); $S=\frac{n}{2}(a+l)$ (1); $S=216$ (1).	1+1+1
18	$d=-6$ (1); $a=50$ (1); $t_{22}=-76$ (1).	1+1+1
	OR: $d=\frac{67-7}{10}=6$ (1); $S_n=\frac{n}{2}[2a+(n-1)d]$ (1); $S_{18}=1044$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lengths form an AP with $d=-2$ (1); Top rung $=80-20\times2=40$ cm (1); $S=\frac{n}{2}(a+l)$ (1); $S=\frac{21}{2}(80+40)$ (1); Total 1260 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $110/100$ (1); Hence a GP with this common ratio (1); Amount after n years $=P(1+\frac{r}{100})^n$ (1); $=40000\times(1.1)^2$ (1); Amount ₹48400 (1).	1+1+1+1+1
20	Lengths form an AP with $d=-8$ (1); Top rung $=200-16\times8=72$ cm (1); $S=\frac{n}{2}(a+l)$ (1); $S=\frac{17}{2}(200+72)$ (1); Total 2312 cm (1).	1+1+1+1+1
	OR: Each year multiplies by $120/100$ (1); Hence a GP with this common ratio (1); Amount after n years $=P(1+\frac{r}{100})^n$ (1); $=25000\times(1.2)^2$ (1); Amount ₹36000 (1).	1+1+1+1+1

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- Using $t_n=ar^n$ for a GP instead of ar^{n-1}.
- Calling a sequence an AP after checking only the first two terms; the difference must be the same for all consecutive terms.
- Taking S_n of an AP to be the nth term, or finding the nth term as S_n instead of S_n-S_{n-1}.
- Sign errors with a negative common difference or ratio, e.g. $t_5=7+4\times(-2)=-1$, not $7+4\times2$.

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